<u>Solution Manual</u>

Analysis and Design of Elementary MOS Amplifier Stages

- First Edition -
Chapter 3

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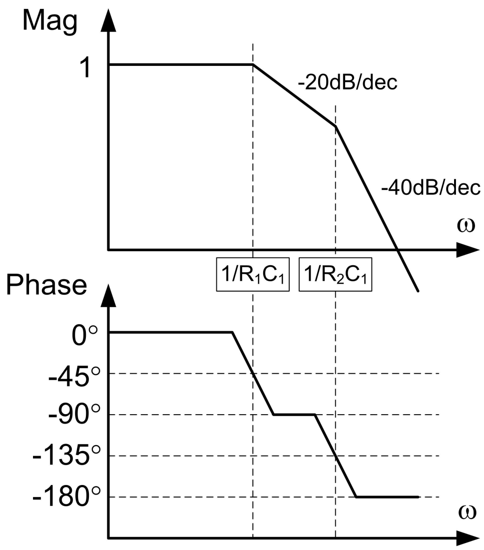
September 2013

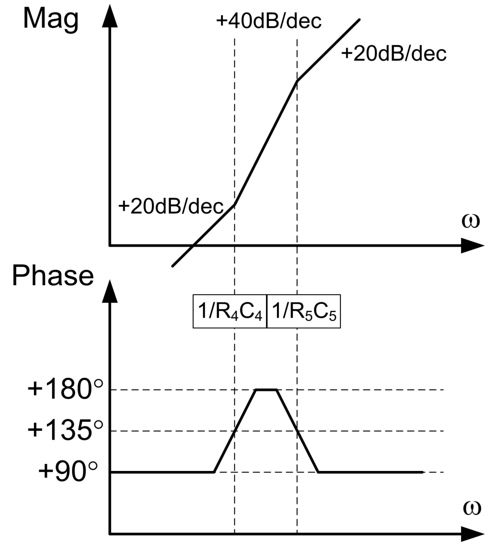
restart : *with*(*Units*[*Standard*]) : *with*(*plots*) : *interface*(*imaginaryunit* = *j*) : **read** "Parameters.mpl" :

P3.1

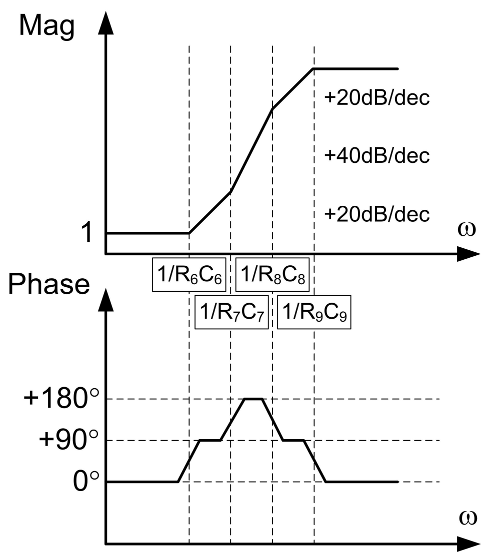
(a)



(b)



(c)



P3.2

$K := 500 : \; z_1 := -2\,\pi\cdot 10[\![kHz]\!] : \; z_2 := -2\,\pi\cdot 1[\![MHz]\!] :$

$p_1 := -2\,\pi\cdot 100[\![kHz]\!] : \; p_2 := -2\,\pi\cdot 10[\![MHz]\!] : p_3 := -2\,\pi\cdot 100[\![MHz]\!] :$

(a) The s-domain function is given below, where the p_i, z_i are negative numbers as listed above:

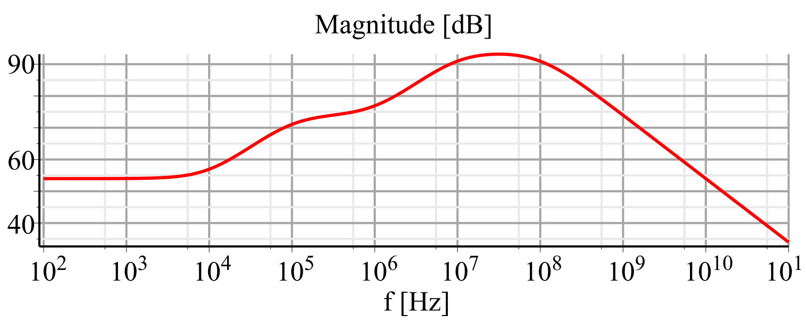
$$H2a := s \rightarrow K\cdot \frac{\left(1-\frac{s[\![Hz]\!]}{z_1}\right)\cdot\left(1-\frac{s[\![Hz]\!]}{z_2}\right)}{\left(1-\frac{s[\![Hz]\!]}{p_1}\right)\cdot\left(1-\frac{s[\![Hz]\!]}{p_2}\right)\cdot\left(1-\frac{s[\![Hz]\!]}{p_3}\right)} :$$

(b) The Bode plot is best constructed using numerical evaluation. We substitute s = j2πf and compute the magnitude and phase as shown below.

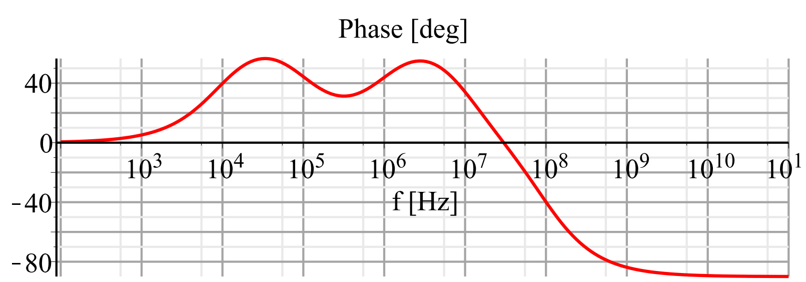
$Mag := x \rightarrow 20\cdot \log_{10}(|x|):$

$Phase := x \rightarrow \frac{180}{\pi}\cdot \mathrm{argument}(x) :$

$semilogplot(Mag(H2a(j\cdot 2\cdot\pi\cdot f)), f = 10^2 .. 10^{11}$, *gridlines*, *title* = "Magnitude [dB]", *labels* = ["f [Hz]", ""])



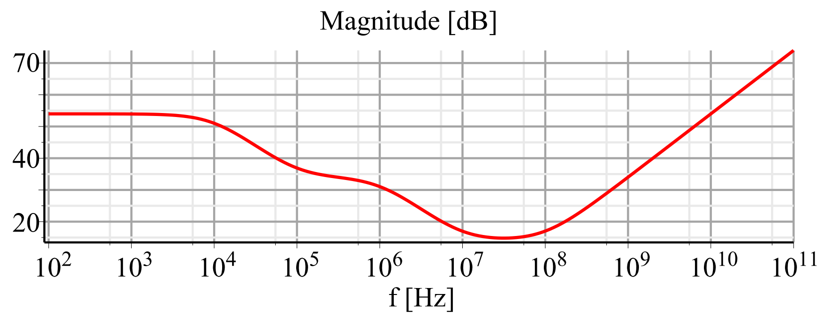
$semilogplot(Phase(H2a(j\cdot 2\cdot\pi\cdot f)), f = 10^2 .. 10^{11}$, *gridlines*, *title* = "Phase [deg]", *labels* = ["f [Hz]", ""])



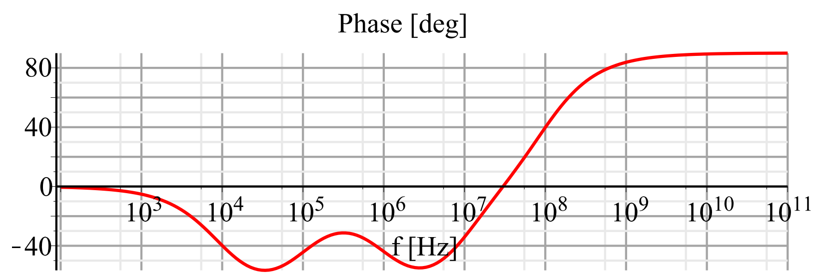
(c) The new s-domain function and corresponding Bode plot is given below:

$$H2c := s \rightarrow K \cdot \frac{\left(1 - \frac{s[\![Hz]\!]}{p_1}\right) \cdot \left(1 - \frac{s[\![Hz]\!]}{p_2}\right) \cdot \left(1 - \frac{s[\![Hz]\!]}{p_3}\right)}{\left(1 - \frac{s[\![Hz]\!]}{z_1}\right) \cdot \left(1 - \frac{s[\![Hz]\!]}{z_2}\right)} :$$

$semilogplot(Mag(H2c(j \cdot 2 \cdot \pi \cdot f)), f = 10^2 .. 10^{11}, gridlines, title =$ "Magnitude [dB]", $labels =$ ["f [Hz]", ""])



$semilogplot(Phase(H2c(j \cdot 2 \cdot \pi \cdot f)), f = 10^2 .. 10^{11}, gridlines, title =$ "Phase [deg]", $labels =$ ["f [Hz]", ""])



P3.3

We begin by deriving the s-domain transfer function of the circuit. First, note that

$$i_o := -\frac{v_o}{R_2} : \quad \text{and}$$

$$v_o := i_s \cdot \left(\frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + s \cdot C} \right) :$$

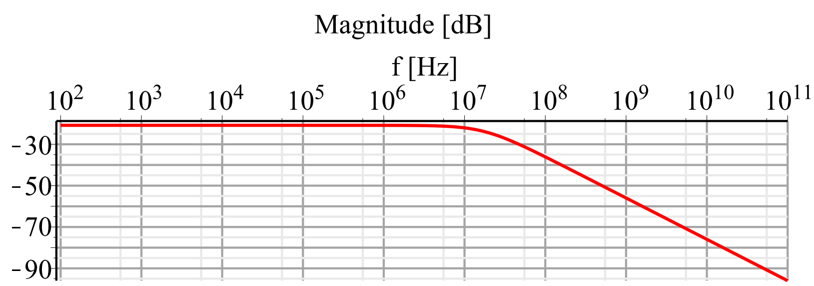
After substituting the second expression into the first, we can write the transfer function H(s) = i_o/i_s as shown below. Note that we purposely isolated the constant term from the frequency dependent part.

$$H3 := s \rightarrow -\frac{R_1}{R_1 + R_2} \cdot \frac{1}{\left(1 + s \cdot C \cdot \frac{R_1 \cdot R_2}{R_1 + R_2}\right)} :$$

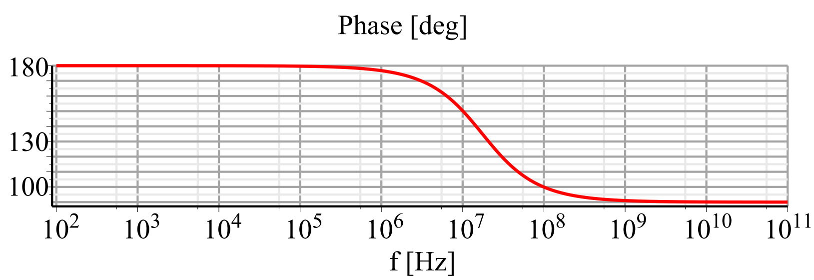
The Bode plots for the different combinations of component values follow below.

(a) $R_1 := 10[\![k\Omega]\!] : R_2 := 100[\![k\Omega]\!] : C := 1[\![pF]\!] :$

$semilogplot\left(Mag\left(H3\left(j \cdot 2 \cdot \pi \cdot f \cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Magnitude [dB]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$

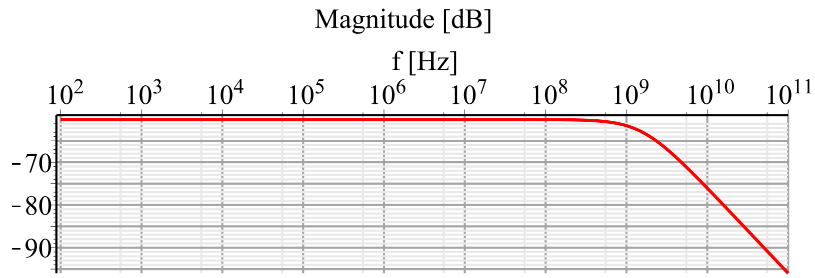


$semilogplot\left(Phase\left(H3\left(j \cdot 2 \cdot \pi \cdot f \cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Phase [deg]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$

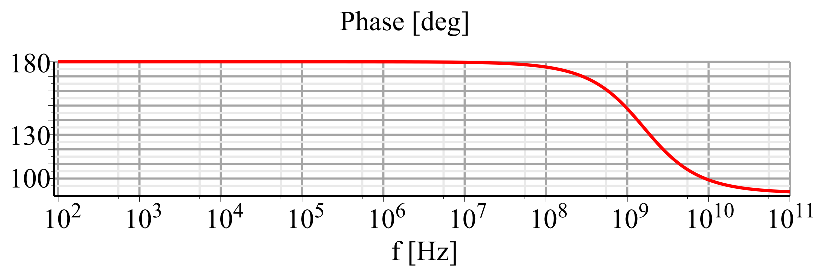


(b) $R_1 := 0.1[\![k\Omega]\!] : R_2 := 100[\![k\Omega]\!] : C := 1[\![pF]\!] :$

$semilogplot\left(Mag\left(H3\left(j\cdot 2\cdot\pi\cdot f\cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Magnitude [dB]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$



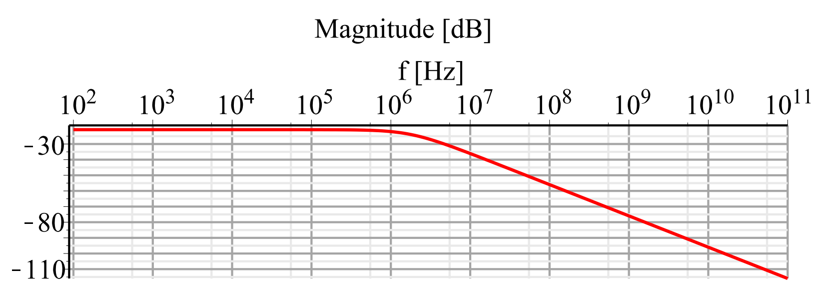
$semilogplot\left(Phase\left(H3\left(j\cdot 2\cdot\pi\cdot f\cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Phase [deg]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$



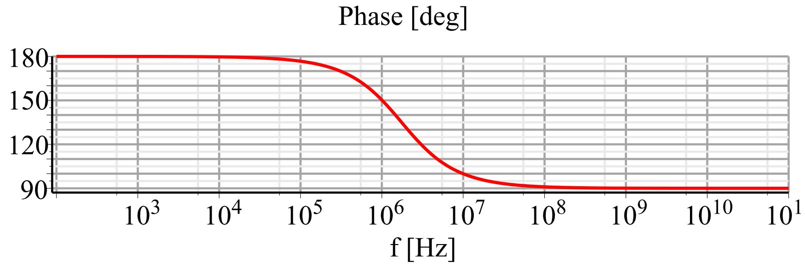
Observations:
1) Lowering R_1 reduces the low-frequency gain (since R_1 and R_2 form a current divider).
2) The pole is shifted to higher frequencies, since the time constant reduces for smaller R_1.

(c) $R_1 := 10[\![k\Omega]\!] : R_2 := 100[\![k\Omega]\!] : C := 10[\![pF]\!] :$

$semilogplot\left(Mag\left(H3\left(j\cdot 2\cdot\pi\cdot f\cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Magnitude [dB]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$



$semilogplot\left(Phase\left(H3\left(j\cdot 2\cdot\pi\cdot f\cdot [\![Hz]\!]\right)\right), f = 10^2 .. 10^{11}, gridlines, title = \text{"Phase [deg]"}, labels = [\text{"f [Hz]"}, \text{""}]\right)$



Observations:

1) The low-frequency gain is unaffected relative to case (a).

2) Relative to case (a), the pole is shifted to lower frequencies, since the time constant increases for larger C.

P3.4

(a) $V_{DB} := 2.5 [\![V]\!] : V_{SB} := 0 [\![V]\!] : I_D := 500 [\![uA]\!] : L := 2 [\![um]\!] : W := 20 [\![um]\!] :$

Note that compared to Ex. 3-5, the channel length L is doubled. First compute all capacitances:

$$C_{gs} := 1. \cdot \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 7.133\ 10^{-14}\ [\![F]\!]$$

$$C_{gd} := W \cdot C_{ov} = 1.000\ 10^{-14}\ [\![F]\!]$$

$$AD := 1. \cdot W \cdot L_{diff} = 6.000\ 10^{-11}\ [\![m^2]\!] PD := W + 2 \cdot L_{diff} = 26\ [\![um]\!]$$

$$C_{db} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 1.161\ 10^{-14}\ [\![F]\!]$$

$AS := AD : PS := PD :$

$$C_{sb} := \frac{C_{Jn} \cdot AS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} = 1.900\ 10^{-14}\ [\![F]\!]$$

Now compute the transconductance:

$$g_m := \sqrt{2. \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot I_D} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 0.707\ [\![mS]\!]$$

The transit frequency is:

$$f_T := \frac{1}{2 \cdot 3.142} \cdot \frac{g_m}{C_{gs} + C_{gd}} = 1.384\ 10^{9} \left[\!\left[\frac{1}{s}\right]\!\right]$$

Compared to the result of Example 3-5 ($f_{TEx} := 3.14 [\![GHz]\!] :$), the transit frequency has changed by the following factor:

$$\frac{f_T}{f_{TEx}} = 0.441$$

Considering only intrinsic capacitance, the above ratio would be 0.25, since f_T scales to first order with $1/L^2$ (Eq. (3.25)). However, since the extrinsic capacitance component does not scale with L, we see a smaller degradation.

(b) $L := 1 [\![um]\!] : W := 40 [\![um]\!] :$

Compared to Ex. 3-5, W is doubled. First compute all capacitances:

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 8.133\ 10^{-14}\ [[F]]$$

$$C_{gd} := W \cdot C_{ov} = 2.000\ 10^{-14}\ [[F]]$$

$$AD := 1. \cdot W \cdot L_{diff} = 1.200\ 10^{-10}\ [[m^2]] \quad PD := W + 2 \cdot L_{diff} = 46\ [[um]]$$

$$C_{db} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 2.126\ 10^{-14}\ [[F]]$$

$$AS := AD : PS := PD :$$

$$C_{sb} := \frac{C_{Jn} \cdot AS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} = 3.500\ 10^{-14}\ [[F]]$$

Now compute the transconductance:

$$g_m := \sqrt{2. \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot I_D} = 0.001\ [[S]] \xrightarrow{\text{replace units}} 1.414\ [[mS]]$$

The transit frequency is:

$$f_T := \frac{1}{2 \cdot 3.142} \cdot \frac{g_m}{C_{gs} + C_{gd}} = 2.221\ 10^9 \left[\left[\frac{1}{s}\right]\right]$$

Compared to the result of Example 3-5, the transit frequency has changed by the following factor:

$$\frac{f_T}{f_{TEx}} = 0.707$$

The transit frequency has changed by a factor of $\frac{1}{\sqrt{2}}$. The transconductance has increased by $\sqrt{2}$, but all capacitances have doubled.

(c) $I_D := 1000[[uA]] : L := 1[[um]] : W := 40[[um]] :$

Compared to Ex. 3-5, I_D and W are doubled. The capacitances are the same as in (b). The transconductance is:

$$g_m := \sqrt{2. \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot I_D} = 0.002\ [[S]] \xrightarrow{\text{replace units}} 2.000\ [[mS]]$$

The transit frequency is:

$$f_T := \frac{1}{2 \cdot 3.142} \cdot \frac{g_m}{C_{gs} + C_{gd}} = 3.141\ 10^9 \left[\left[\frac{1}{s}\right]\right]$$

Compared to the result of Example 3-5, the transit frequency has stayed the same.

$$\frac{f_T}{f_{TEx}} = 1.000$$

The transit frequency of a MOSFET stays constant as long as L and the current density of the transistor are unchanged. Changing W and I_D by the same factor does not affect f_T.

P3.5

(a) $V_{DB} := 2.5[\![V]\!] : V_{SB} := 0[\![V]\!] : I_D := 500[\![uA]\!] : L := 1[\![um]\!] : W := 20[\![um]\!] :$

C_{gs} and C_{gd} are exactly the same as in Ex. 3-5:

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 4.067\ 10^{-14}\ [\![F]\!]$$
$$C_{gd} := W \cdot C_{ov} = 1.000\ 10^{-14}\ [\![F]\!]$$

The junction capacitances are larger due the larger C_J parameters for the PMOS (see Table 3-1):

$$AD := 1. \cdot W \cdot L_{diff} = 6.000\ 10^{-11}\ [\![m^2]\!] \qquad PD := W + 2 \cdot L_{diff} = 26\ [\![um]\!]$$
$$C_{db} := \frac{C_{Jp} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWp} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 1.537\ 10^{-14}\ [\![F]\!]$$

$$AS := AD : PS := PD :$$
$$C_{sb} := \frac{C_{Jp} \cdot AS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} + \frac{C_{JSWp} \cdot PS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} = 2.710\ 10^{-14}\ [\![F]\!]$$

Now compute the transconductance:

$$g_m := \sqrt{2. \cdot \mu p C_{ox} \cdot \frac{W}{L} \cdot I_D} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 0.707\ [\![mS]\!]$$

The transit frequency is:

$$f_T := \frac{1}{2 \cdot 3.142} \cdot \frac{g_m}{C_{gs} + C_{gd}} = 2.221\ 10^{9}\ \left[\!\left[\frac{1}{s}\right]\!\right]$$

Compared to the result of Ex. 3-5 ($f_{TEx} := 3.14[\![GHz]\!] :$), the transit frequency has changed by the following factor:

$$\frac{f_T}{f_{TEx}} = 0.707$$

The transit frequency is reduced by a factor of $\frac{1}{\sqrt{2}}$. This is the square root of the PMOS/NMOS mobility ratio.

P3.6

$V_{DB} := 2.5 [\![V]\!] : W := 100 [\![um]\!] :$

$AD := 1. \cdot W \cdot L_{diff} = 3.000\ 10^{-10} [\![m^2]\!] \quad PD := W + 2 \cdot L_{diff} = 106\ [\![um]\!]$

$$C_{db25} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 5.022\ 10^{-14} [\![F]\!]$$

Now repeat for $V_{DB} := 1\ [\![V]\!] :$

$$C_{db1} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 6.264\ 10^{-14} [\![F]\!]$$

Now repeat for $V_{DB} := 4\ [\![V]\!] :$

$$C_{db4} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 4.371\ 10^{-14} [\![F]\!]$$

The corresponding ratios are:

$$\frac{C_{db1}}{C_{db25}} = 1.247$$

$$\frac{C_{db4}}{C_{db25}} = 0.870$$

This means that the drain-bulk capacitance changes by about +25% to -13% when the voltage is swept from the center to within 1V of the supply rails.

P3.7

$W_1 := 10[\![um]\!] : L_1 := 1[\![um]\!] : W_2 := 10[\![um]\!] : L_2 := 10[\![um]\!] : V_S := 2[\![V]\!] : V_{DD} := 5\ [\![V]\!] :$

The total capacitance at node v_o is

$$C := W_2 \cdot L_2 \cdot C_{ox} + \frac{1}{2} \cdot W_1 \cdot L_1 \cdot C_{ox} = 2.415\ 10^{-13}\ [\![F]\!]$$

The on-resistance of M_1 is

$$R := \frac{1}{\mu n C_{ox} \cdot \frac{W_1}{L_1} \cdot (V_{DD} - V_S - V_{T0n})} = 800.000\ [\![\Omega]\!]$$

The bandwidth of the circuit is therefore

$$f_{3\,dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R \cdot C} = 8.237\ 10^{8} \left[\!\!\left[\frac{1}{s} \right]\!\!\right]$$

P3.8

$g_m := 1[\![mS]\!] : C_{gs} := 40.7[\![fF]\!] : C_{gd} := 10[\![fF]\!] : C_{db} := 11.6\ [\![fF]\!] : R_{out} := 5[\![k\Omega]\!] : R_s := 50[\![k\Omega]\!] :$

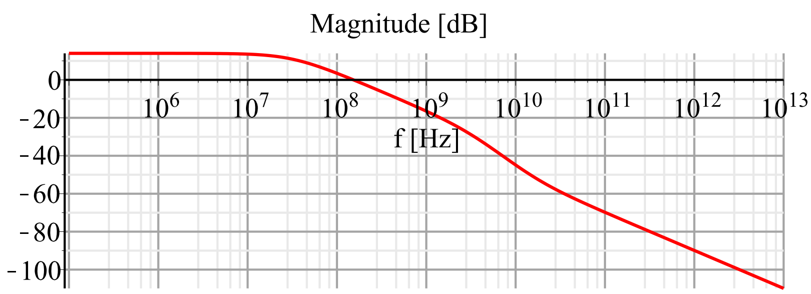
$$a_1 := -1.\cdot\frac{C_{gd}}{g_m} = -1.000\ 10^{-11}\ [\![s]\!]$$

$$b_1 := R_s\cdot\left(C_{gs}+C_{gd}\right) + R_{out}\cdot\left(C_{db}+C_{gd}\right) + g_m\cdot R_{out}\cdot R_s\cdot C_{gd} = 5.143\ 10^{-9}\ [\![s]\!]$$

$$b_2 := R_s\cdot R_{out}\cdot\left(C_{gs}\cdot C_{gd} + C_{gs}\cdot C_{db} + C_{gd}\cdot C_{db}\right) = 2.488\ 10^{-19}\ [\![s^2]\!]$$

$$H8 := s \rightarrow -g_m\cdot R_{out}\cdot\frac{1 + a_1\cdot s[\![Hz]\!]}{1 + b_1\cdot s[\![Hz]\!] + b_2\cdot s^2[\![Hz]\!]^2} :$$

semilogplot(*Mag*(*H8*($j\cdot 2\cdot\pi\cdot f$)), $f = 10^5 .. 10^{13}$, *gridlines*, *title* = "Magnitude [dB]", *labels* = ["f [Hz]", ""])



In the above plot, note the far out location of the second pole (between 10^9 and 10^{10}) and the zero (near 10^{10}). The exact value of the 3dB frequency [in MHz] is computed below.

$$f_{3\,dB} := fsolve\left(\left|\frac{1 + a_1\cdot\left(j\cdot 2\cdot\pi\cdot f\cdot[\![MHz]\!]\right)}{1 + b_1\cdot\left(j\cdot 2\cdot\pi\cdot f\cdot[\![\mathrm{MHz}]\!]\right) + b_2\cdot\left(j\cdot 2\cdot\pi\cdot f\cdot[\![MHz]\!]\right)^2}\right| = \frac{1}{\sqrt{2}}, f, real\right) = 31.240$$

The numbers obtained in Ex. 3-7 ($f_{3\,dBMiller} := 31.61 : f_{3\,dBDominant} := 30.95 :$) are very close to this exact result. This is to be expected, since the second pole and the high frequency zero have negligible impact on the transfer function magnitude near the 3dB point.

$$100\cdot\frac{f_{3\,dBMiller} - f_{3\,dB}}{f_{3\,dB}} = 1.185\ \%$$

Note that this error percentage is expected to be positive, since the Miller approximation overestimates the bandwidth.

$$100 \cdot \frac{f_{3\,dBDominant} - f_{3\,dB}}{f_{3\,dB}} = -0.928\ \%$$

Note that this error percentage is expected to be negative, since the dominant pole approximation uses $1/b_1$ as the bandwidth estimate and is therefore conservative (just like OCT analysis).

P3.9

$g_m := 1[\![mS]\!] : C_{gs} := 40.7[\![fF]\!] : C_{gd} := 10[\![fF]\!] : C_{db} := 11.6\ [\![fF]\!] : R_{out} := 5[\![k\Omega]\!] : R_s$
$:= 50[\![k\Omega]\!] :$

$b_1 := R_s\cdot(C_{gs} + C_{gd}) + R_{out}\cdot(C_{db} + C_{gd}) + g_m\cdot R_{out}\cdot R_s\cdot C_{gd} = 5.143\ 10^{-9}\ [\![s]\!]$
$b_2 := R_s\cdot R_{out}\cdot(C_{gs}\cdot C_{gd} + C_{gs}\cdot C_{db} + C_{gd}\cdot C_{db}) = 2.488\ 10^{-19}\ [\![s^2]\!]$

We can estimate the nondominant pole frequency using Eq. (3.58):

$$f_{p2} := \frac{1}{2\cdot 3.142}\cdot\frac{b_1}{b_2} = 3.290\ 10^9\ \left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

Alternatively, we can try to use the approximation given in Eq. (3.59):

$$f_{p2} := \frac{1}{2\cdot 3.142}\cdot\left(\frac{1}{R_{out}\cdot C_{db}} + \frac{1}{R_s\cdot C_{gs}} + \frac{g_m}{C_{db}}\cdot\frac{C_{gd}}{C_{gs}}\right) = 6.193\ 10^9\ \left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

This approximation is not accurate in this case, since $C_{gd} << C_{db}$ is not satisfied. We can compute the exact roots of the denominator polynomial to make sure that the first result is indeed correct:

$r := fsolve\left(1 + b_1\cdot s\cdot [\![MHz]\!] + b_2\cdot s^2\cdot [\![MHz]\!]^2 = 0, s\right) = -20476.581, -196.303$

$$f_{p2} := -\frac{r_1\cdot [\![MHz]\!]}{2\cdot evalf(\pi)} = 3258.949\ [\![MHz]\!]$$

P3.10

For $A_{VM} = 1$, the voltages at the two capacitor terminals are equal. Consequently, no current flows in the capacitor and it seen as an infinite impedance from the input side (no current implies an open circuit).

P3.11

If we were to use Eq. (3.56) to estimate the bandwidth for Example 3-8, we would get the same answer as already computed in Example 3-7, i.e. f_{3dB} = 31.61 MHz. The is clearly wrong in light of the OCT estimate we computed in Example 3-8 (f_{3dB} = 2.89 MHz). The issue is that for C_L = 1 pF, the pole frequency in Eq. (3.54) is not larger than the 3-dB bandwidth we are trying to estimate (see calculation below). The third step in applying the Miller approximation, given on page 72, is immensely important when using the Miller approximation.

$C_L := 10[\![pF]\!] : C_{gd} := 10[\![fF]\!] : C_{db} := 11.6\,[\![fF]\!] : R_{out} := 5[\![k\Omega]\!] :$

$$\frac{1}{2\cdot 3.142}\cdot\frac{1}{R_{out}\cdot\left(C_{db}+C_L+C_{gd}\right)} = 3.176\;10^6\left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

This number is smaller than 31.61 MHz and using the Miller approximation is therefore invalid.

P3.12

$W := 100[\![um]\!] : L := 2[\![um]\!] : I_B := 3[\![mA]\!] : V_B := 2.5[\![V]\!] : C_L : 100[\![fF]\!] : R_{\mathrm{D}} := 1[\![k\Omega]\!] : R_s$
$:= 10[\![k\Omega]\!] :$

(a)

$$V_{OV} := \sqrt{\frac{2.\cdot I_B}{\mu n C_{ox}\cdot \frac{W}{L}}} = 1.549\ [\![V]\!]$$

$$V_S := V_{T0n} + V_{OV} = 2.049\ [\![V]\!]$$

(b) $g_m := \dfrac{2\cdot I_B}{V_{OV}} = 0.004\ [\![S]\!] \xrightarrow{\text{replace units}} 3.873\ [\![mS]\!]$

$$C_{gs} := \frac{2}{3}\cdot W\cdot L\cdot C_{ox} + W\cdot C_{ov} = 3.567\ 10^{-13}\ [\![F]\!]$$

$$C_{gd} := W\cdot C_{ov} = 5.000\ 10^{-14}\ [\![F]\!]$$

$AD := 1.\cdot\ W\cdot L_{diff} = 3.000\ 10^{-10}\ [\![m^2]\!] \qquad PD := W + 2\cdot L_{diff} = 106\ [\![um]\!]$

$$C_{db} := \frac{C_{Jn}\cdot AD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} + \frac{C_{JSWn}\cdot PD}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} = 4.371\ 10^{-14}\ [\![F]\!]$$

$AS := AD : PS := PD :$

$$C_{sb} := \frac{C_{Jn}\cdot AS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} + \frac{C_{JSWn}\cdot PS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} = 8.300\ 10^{-14}\ [\![F]\!]$$

(c) $C_{gs,\ intrinsic} := \dfrac{2}{3}\cdot W\cdot L\cdot C_{ox} = 3.067\ 10^{-13}\ [\![F]\!]$

$$f_{3\ dB,\ intrinsic} := \frac{1}{2\cdot 3.142}\cdot\frac{1}{R_s\cdot C_{gs,\ intrinsic}} = 5.189\ 10^{7}\left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

(d) $f_{3\ dB,\ Miller} := \dfrac{1}{2\cdot 3.142}\cdot\dfrac{1}{R_s\cdot\left(C_{gs} + \left(1 + g_m\cdot R_{\mathrm{D}}\right)\cdot C_{gd}\right)} = 2.651\ 10^{7}\left[\!\!\left[\dfrac{1}{s}\right]\!\!\right]$

(e) $\tau_{gso} := R_s\cdot C_{gs} = 3.567\ 10^{-9}\ [\![s]\!]$

$$\tau_{\mathrm{dbo}} := R_{\mathrm{D}}\cdot\left(C_{db} + C_L\right) = 1.004\ 10^{-8}\ [\![s]\!]$$

$$\tau_{gdo} := \left(R_s + R_{\mathrm{D}} + g_m\cdot R_s\cdot R_{\mathrm{D}}\right)\cdot C_{gd} = 2.486\ 10^{-9}\ [\![s]\!]$$

$$f_{3\ dB,\ OCT} := \frac{1}{2\cdot 3.142}\cdot\frac{1}{\tau_{gso} + \tau_{\mathrm{dbo}} + \tau_{gdo}} = 9.886\ 10^{6}\left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

As we can see from these results, using only intrinsic gate capacitance or applying the Miller approximation gives overly optimistic results for this circuit. For reference (not required for this problem), the exact bandwidth of the circuit is given below. As expected, the OCT estimate is sufficiently accurate for first-cut design.

$$a_1 := -1.\cdot\frac{C_{gd}}{g_m} = -1.291\ 10^{-11}\ [\![s]\!]$$

$$b_1 := R_s\cdot(C_{gs}+C_{gd}) + R_\mathrm{D}\cdot(C_{db}+C_L+C_{gd}) + g_m\cdot R_\mathrm{D}\cdot R_s\cdot C_{gd} = 1.610\ 10^{-8}\ [\![s]\!]$$

$$b_2 := R_s\cdot R_\mathrm{D}\cdot(C_{gs}\cdot C_{gd} + C_{gs}\cdot(C_{db}+C_L) + C_{gd}\cdot(C_{db}+C_L)) = 4.102\ 10^{-17}\ [\![s^2]\!]$$

$$f_{3\,dB} := -fsolve\left(\left|\frac{1+a_1\cdot(j\cdot 2\cdot\pi\cdot f\cdot[\![MHz]\!])}{1+b_1\cdot(j\cdot 2\cdot\pi\cdot f\cdot[\![\mathrm{MHz}]\!])+b_2\cdot(j\cdot 2\cdot\pi\cdot f\cdot[\![MHz]\!])^2}\right| = \frac{1}{\sqrt{2}}, f, real\right)[\![MHz]\!] = 11.666\ [\![MHz]\!]$$

The percent error in the OCT estimate is:

$$100\cdot\frac{f_{3\,dB,\,OCT}-f_{3\,dB}}{f_{3\,dB}} = -15.261$$

P3.13

(a) $\omega_{3\,dB,\,OCT} := \dfrac{1}{3 \cdot R \cdot C}$

(b) $\left| \dfrac{1}{\left(1 + j \cdot \omega_{3\,dB} \cdot R \cdot C\right)^{3}} \right| = \dfrac{1}{\sqrt{2}}$

$$\left|1 + j \cdot \omega_{3\,dB} \cdot R \cdot C\right| = 2^{\frac{1}{3 \cdot 2}}$$

$$1 + \left(\omega_{3\,dB} \cdot R \cdot C\right)^{2} = 2^{\frac{1}{3}}$$

$$\omega_{3\,dB} := \frac{1}{RC}\sqrt{2^{\frac{1}{3}} - 1}$$

(c) The percent error in the OCT estimate is:

$$100 \cdot \frac{\omega_{3\,dB,\,OCT} - \omega_{3\,dB}}{\omega_{3\,dB}}$$

$$100 \cdot \frac{\frac{1}{3} - \sqrt{2^{\frac{1.}{3}} - 1}}{\sqrt{2^{\frac{1.}{3}} - 1}} =$$

P3.14

$$\omega_{3\,dB,\,OCT} := \frac{\omega_p}{n}$$

$$\left| \frac{1}{\left(1 + j\cdot\frac{\omega_{3\,dB}}{\omega_p}\right)^n} \right| = \frac{1}{\sqrt{2}}$$

$$\left| 1 + j\cdot\frac{\omega_{3\,dB}}{\omega_p} \right| = 2^{\frac{1}{n\cdot 2}}$$

$$1 + \left(\frac{\omega_{3\,dB}}{\omega_p}\right)^2 = 2^{\frac{1}{n}}$$

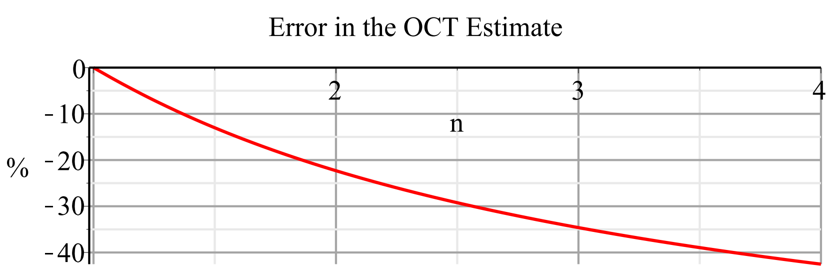
$$\omega_{3\,dB} := \omega_p\sqrt{2^{\frac{1}{n}} - 1}$$

The percent error in the OCT estimate is:

$$100\cdot\frac{\omega_{3\,dB,\,OCT} - \omega_{3\,dB}}{\omega_{3\,dB}}$$

$$f := n \rightarrow 100\cdot\frac{\frac{1}{n} - \sqrt{2^{\frac{1.}{n}} - 1}}{\sqrt{2^{\frac{1.}{n}} - 1}} :$$

$plot(f(n), n = 1..4, gridlines, title = \text{"Error in the OCT Estimate"}, labels = [\text{"n"}, \text{"\%"}])$



P3.15

g_m :='g_m': R_s :='R_s': R_D :='R_D': C_{gs} :='C_{gs}': C_L :='C_L': A_{v0} :='A_{v0}': V_{OV} :='V_{OV}': W :='W': L :='L': I_D :=
'I_D': $\omega_{3\,dB}$:='$\omega_{3\,dB}$': μ_n :='μ_n':

We begin by writing the OCT bandwidth estimate for the circuit. Given that $R_s = 2R_D$, we have:

$$BW := \frac{1}{2\cdot R_D\cdot C_{gs} + R_D\cdot C_L} :$$

Next, we replace all unknowns using knows parameters and circuit specifications.

$$R_D := \frac{|A_{v0}|}{g_m} :$$

$$C_{gs} := \frac{2}{3}\cdot W\cdot L\cdot C_{oxn} :$$

$$W := \frac{I_D}{\frac{1}{2}\cdot\mu_n\cdot C_{oxn}\cdot\frac{1}{L}\cdot V_{OV}^2} :$$

$$g_m := \frac{2\cdot I_D}{V_{OV}} :$$

$$BW = \frac{1}{\frac{4}{3}\,\frac{|A_{v0}|\,L^2}{V_{OV}\,\mu_n} + \frac{1}{2}\,\frac{|A_{v0}|\,V_{OV}\,C_L}{I_D}}$$

Now solve for I_D to get the desired expression:

$$solve\left(\omega_{3\,dB} = \mathrm{BW}, I_D\right) = -\frac{3}{2}\,\frac{\omega_{3\,dB}\,|A_{v0}|\,V_{OV}^{\,2}\,C_L\,\mu_n}{4\,\omega_{3\,dB}\,|A_{v0}|\,L^2 - 3\,V_{OV}\,\mu_n}$$

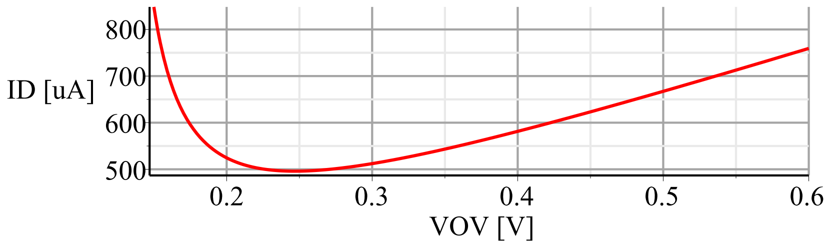
We now plot this expression for the given values.

$$A_{v0} := -4 : \omega_{3\,dB} := 2\cdot 3.142\cdot 80[\![MHz]\!] : C_L := 1[\![pF]\!] : L := 1[\![um]\!] : \mu_n := \frac{\mu n C_{ox}}{C_{ox}} :$$

$$I_D := V_{OV} \rightarrow \frac{1}{2}\cdot\frac{C_L\cdot|A_{v0}|\cdot\omega_{3\,dB}\cdot V_{OV}\cdot[\![V]\!]}{1 - \frac{4}{3}\cdot\frac{L^2}{\mu_n\cdot V_{OV}[\![V]\!]}\cdot|A_{v0}|\cdot\omega_{3\,dB}} :$$

$$plot\left(\frac{I_D(V_{OV})}{[\![uA]\!]}, V_{OV} = 0.15\,..0.6, gridlines, title\right.$$

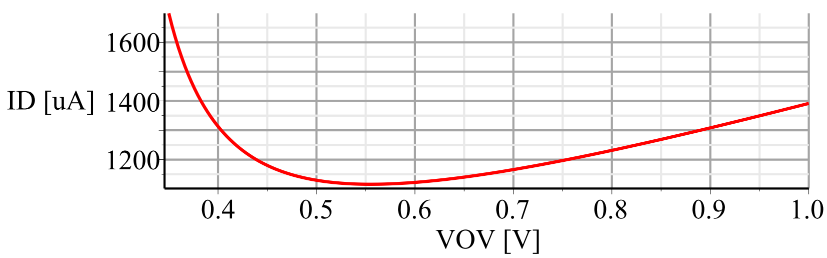
$= \text{"Drain current as a function of gate overdrive for L=1um"}, \mathit{labels} = [\text{"VOV [V]"}, \text{"ID [uA]"}]\Big);$



$$L := 1.5 [\![um]\!] :$$

$$\mathit{plot}\left(\frac{I_D(V_{OV})}{[\![uA]\!]}, V_{OV} = 0.35..1, \mathit{gridlines}, \mathit{title}\right.$$

$$\left. = \text{"Drain current as a function of gate overdrive for L=1um"}, \mathit{labels} = [\text{"VOV [V]"}, \text{"ID [uA]"}]\right)$$



(b) A shorter channel MOSFET contributes less capacitance for a given transconductance (higher transit frequency). This means that for a given desired voltage gain, we can use larger R_D and smaller g_m to achieve the same bandwidth. This leads to a lower drain current requirement.

(c) The drain current expression is of the following form:

$$I_D := \frac{a \cdot V}{1 - \frac{b}{V}}$$

The optimum is found by setting the first derivative of this expression to zero:

$$\mathit{simplify}\left(\frac{d}{dV} \frac{a \cdot V}{1 - \frac{b}{V}}\right) = \frac{a\, V\, (V - 2\, b)}{(V - b)^2}$$

$$L := 1 [\![um]\!] :$$

$$b := \frac{4}{3} \cdot \frac{\omega_{3\,dB} \cdot |A_{v0}| \cdot L^2}{\mu_n} = 0.123 [\![V]\!]$$

$V_{OVopt} := 2 \cdot b \ = 0.247 \ [\![V]\!]$

We can now compute the drain current, device width and R_D for this optimum point:

$$I_{Dopt} := I_D\left(\frac{V_{OVopt}}{[\![V]\!]}\right) = 0.000 \ [\![A]\!] \xrightarrow{\text{replace units}} 496.020 \ [\![uA]\!]$$

$$W_{opt} := \frac{I_{Dopt}}{\frac{1}{2} \ \mu nC_{ox} \cdot \frac{1}{L} \cdot V_{OVopt}^2} = 0.000 \ [\![m]\!] \xrightarrow{\text{replace units}} 326.087 \ [\![um]\!]$$

$$g_{mopt} := \frac{2 \cdot I_{Dopt}}{V_{OVopt}} = 0.004 \ [\![S]\!] \xrightarrow{\text{replace units}} 4.022 \ [\![mS]\!]$$

$$R_{Dopt} := \frac{|A_{v0}|}{g_{mopt}} = 994.589 \ [\![\Omega]\!]$$

(d) The circuit can be simulated using HSpice and the provided netlist "P3.15d_netlist.sp." Below are the obtained results for the DC operating point and AC analyses.

```
*******
 element  0:m1
 model    0:nch.1
 region     Saturati
  id       615.6172u
  ibs        0.
  ibd      -23.8110f
  vgs      747.0000m
  vds        2.3811
  vbs        0.
  vth      500.0000m
  vdsat    247.0000m
  vod      247.0000m
  beta      20.1812m
  gam eff  600.0000m
  gm         4.9848m
  gds       49.7223u
  gmb        1.6719m
  cdtot    435.0746f
  cgtot    842.3564f
  cstot      1.0897p
  cbtot    710.6008f
  cgs      662.8692f
  cgd      165.3805f
```

Note that I_D and g_m of the transistor are larger than designed. This due to channel-length modulation.

```
  ******  ac analysis tnom=  25.000 temp=  25.000 *****
   adc=  4.7215E+00
   freq3db=  3.8756E+07
```

The percent errors in the AC analysis results are:

$$100 \cdot \frac{4.72 - |A_{v0}|}{|A_{v0}|} = 18.000$$

$$100 \cdot \frac{38.75[\![MHz]\!] - 80[\![MHz]\!]}{80[\![MHz]\!]} = -51.563$$

(e) In the calculations below, we use a more accurate OCT estimate that excludes the extrinsic capacitances and also include channel-length modulation. We begin by assuming the same device width as in the first-order result and then step up and down to see if we can reduce the current. The advantage of using W as the sweep parameter is that all capacitances can be computed directly.

Fixed parameters:

$$A_{v0} := -4 : \omega_{3\,dB} := 2 \cdot 3.142 \cdot 80[\![MHz]\!] : C_L := 1[\![pF]\!] : L := 1[\![um]\!] : \mu_n := \frac{\mu n C_{ox}}{C_{ox}} : V_{DS} := 2.5[\![V]\!] :$$

Main variable for iterations:

$$W := 326[\![um]\!] :$$

Design equations:

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 6.629\ 10^{-13}\ [\![F]\!]$$

$$C_{gd} := W \cdot C_{ov} = 1.630\ 10^{-13}\ [\![F]\!]$$

$$AD := 1. \cdot W \cdot L_{diff} = 9.780\ 10^{-10}\ [\![m^2]\!] \qquad PD := W + 2 \cdot L_{diff} = 332\ [\![um]\!]$$

$$C_{db} := \frac{C_{Jn} \cdot AD}{\left(1 + \frac{V_{DS}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PD}{\left(1 + \frac{V_{DS}}{PB}\right)^{MJSW}} = 1.593\ 10^{-13}\ [\![F]\!]$$

$$BW := \frac{1}{2 \cdot R_D \cdot C_{gs} + 2 \cdot R_D \cdot (1 + |A_{v0}|) \cdot C_{gd} + R_D (C_L + C_{db})}$$

$$R_D := \frac{1}{\omega_{3\,dB} \cdot \left(2 \cdot C_{gs} + 2 \cdot (1 + |A_{v0}|) \cdot C_{gd} + C_L + C_{db}\right)} = 483.391\ [\![\Omega]\!]$$

$$g_m := \frac{|A_{v0}|}{R_D} = 0.008\ [\![S]\!] \xrightarrow{\text{replace units}} 8.275\ [\![mS]\!]$$

$$I_D := \frac{g_m^2}{2 \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot \left(1 + \frac{\lambda_n}{L} \cdot V_{DS}\right)} = 0.002\ [\![A]\!] \xrightarrow{\text{replace units}} 1680.332\ [\![uA]\!]$$

$$V_{OV} := \frac{2 \cdot I_D}{g_m} = 0.406\ [\![V]\!]$$

By stepping through W, we can identify a new design point:

W [[um]]	I_D [[uA]]
50	1460
75	1273
100	**1238** (chosen design point)
200	1371
300	1612
326	**1680** (starting point)
350	1744

Even though the initial and final design points lie far apart, the initial first-order analysis is still important because: (1) It gave us analytical insight that there exists a drain current minimum, and (2) it provided us with a starting point for iterations. The parameters for the final design are given below. We include one more refinement and add r_o to the gain calculation using the obtained current estimate.

$$W := 100[\![um]\!] :$$

$$C_{gs} := \frac{2}{3}\cdot W\cdot L\cdot C_{ox} + W\cdot C_{ov} = 2.033\ 10^{-13}\ [\![F]\!]$$

$$C_{gd} := W\cdot C_{ov} = 5.000\ 10^{-14}\ [\![F]\!]$$

$$AD := 1.\cdot\ W\cdot L_{diff} = 3.000\ 10^{-10}\ [\![m^2]\!] \qquad PD := W + 2\cdot L_{diff} = 106\ [\![um]\!]$$

$$C_{db} := \frac{C_{Jn}\cdot AD}{\left(1+\frac{V_{DS}}{PB}\right)^{MJ}} + \frac{C_{JSWn}\cdot PD}{\left(1+\frac{V_{DS}}{PB}\right)^{MJSW}} = 5.022\ 10^{-14}\ [\![F]\!]$$

$$R_D := \frac{1}{\omega_{3\,dB}\cdot\left(2\cdot C_{gs} + 2\cdot\left(1+|A_{v0}|\right)\cdot C_{gd} + C_L + C_{db}\right)} = 1016.500\ [\![\Omega]\!]$$

$$r_o := \frac{1+\frac{\lambda_n}{L}\cdot V_{DS}}{\frac{\lambda_n}{L}\cdot 1238[\![uA]\!]} = 10096.931\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 10.097\ [\![kOmega]\!]$$

$$g_m := \frac{|A_{v0}|}{\left(\frac{1}{R_D}+\frac{1}{r_o}\right)^{-1}} = 0.004\ [\![S]\!] \xrightarrow{\text{replace units}} 4.331\ [\![mS]\!]$$

$$I_D := \frac{g_m^2}{2\cdot \mu nC_{ox}\cdot\frac{W}{L}\cdot\left(1+\frac{\lambda_n}{L}\cdot V_{DS}\right)} = 0.002\ [\![A]\!] \xrightarrow{\text{replace units}} 1500.765\ [\![uA]\!]$$

$$V_{OV} := \frac{2\cdot I_D}{g_m} = 0.693\ [\![V]\!]$$

(f) The updated simulation results are:

```
element   0:m1
model     0:nch.1
region      Saturati
 id           1.5007m
 ibs          0.
 ibd        -24.9929f
 vgs          1.1930
 vds          2.4993
 vbs          0.
 vth        500.0000m
 vdsat      693.0000m
 vod        693.0000m
 beta         6.2496m
 gam eff    600.0000m
 gm           4.3310m
 gds        120.0622u
 gmb          1.4527m
 cdtot      133.8136f
 cgtot      255.7210f
 cstot      336.3341f
 cbtot      217.6676f
 cgs        203.3341f
 cgd         50.7665f
```

Note that all small-signal parameters are in nearly perfect agreement with the above calculations.

```
 ******  ac analysis tnom=  25.000 temp=  25.000 *****
  adc=  3.9253E+00
  freq3db=  9.4254E+07
```

The percent errors in the AC analysis results are:

$$100 \cdot \frac{3.93 - |A_{v0}|}{|A_{v0}|} = -1.750$$

$$100 \cdot \frac{94.25 [\![MHz]\!] - 80 [\![MHz]\!]}{80 [\![MHz]\!]} = 17.813$$

The bandwidth spec is now comfortably met. The margin is due to the conservative nature of the employed OCT bandwidth estimate. Compared to our initial design from part (c), we had to invest quite a bit of extra current:

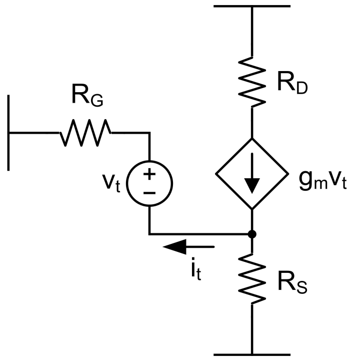
$$100 \cdot \frac{1.5 [\![mA]\!] - 0.496 [\![mA]\!]}{0.496 [\![mA]\!]} = 202.419$$

The extra current is essentially needed to overcome the extrinsic capacitances and thus highlights their significance.

P3.16

$g_m :=' g_m ': R_S :=' R_S ': R_D :=' R_D ':$

To find R_{gs}, we apply a test voltage source across the v_{gs} port and write KCL at the source node of the transistor. As a second equation, we write KVL around the loop involving R_G.



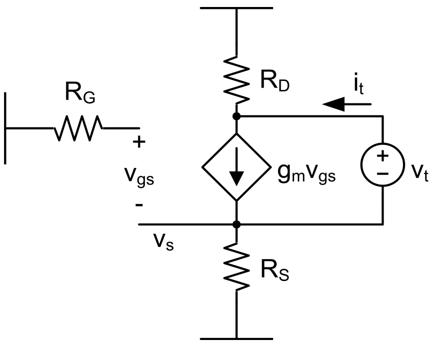
$$Eq1 := 0 = \frac{v_s}{R_S} + i_t - g_m \cdot v_t :$$

$$Eq2 := v_s = i_t \cdot R_G - v_t :$$

$$solve(\{Eq1, Eq2\}, [v_t, v_s]) = \left[\left[v_t = \frac{i_t (R_G + R_S)}{1 + g_m R_S}, v_s = \frac{i_t R_S (R_G g_m - 1)}{1 + g_m R_S}\right]\right]$$

$$\frac{v_t}{i_t} := \frac{R_G + R_S}{1 + g_m R_S}$$

To find R_{ds}, we apply a test voltage source across the v_{ds} port and write KCL at the two nodes of the circuit.



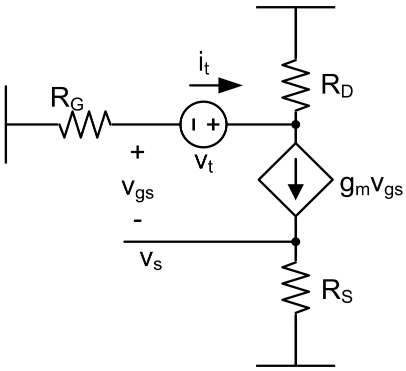
$$Eq1 := 0 = \frac{v_s}{R_S} + i_t + g_m \cdot v_s :$$

$$Eq2 := 0 = -i_t - g_m \cdot v_s + \frac{v_s + v_t}{R_\mathrm{D}} :$$

$$solve(\{Eq1, Eq2\}, [v_t, v_s]) = \left[\left[v_t = \frac{i_t (R_D + R_S)}{1 + g_m R_S}, v_s = -\frac{i_t R_S}{1 + g_m R_S}\right]\right]$$

$$\frac{v_t}{i_t} := \frac{R_D + R_S}{1 + g_m R_S} :$$

To find R_{gd}, we apply a test voltage source across the v_{ds} port and write KCL at the source and drain and gate nodes of the circuit.



$$Eq1 := 0 = \frac{v_s}{R_S} - g_m \cdot (v_g - v_s) :$$

$$Eq2 := 0 = -i_t + g_m \cdot (v_g - v_s) + \frac{v_g + v_t}{R_\mathrm{D}} :$$

$$Eq3 := 0 = \frac{v_g}{R_G} + i_t :$$

$$solve(\{Eq1, Eq2, Eq3\}, [v_t, v_g, v_s]) =$$

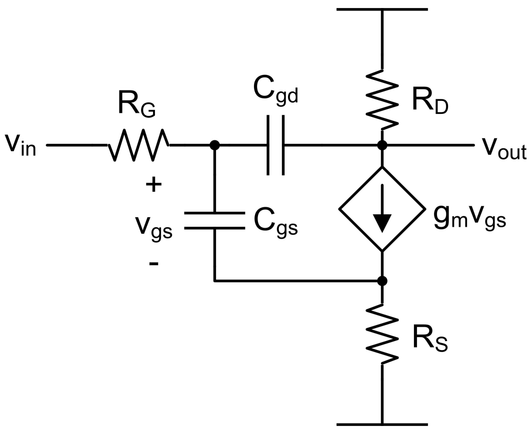
$$\left[\left[v_t = \frac{i_t (R_D + g_m R_D R_G + R_G + g_m R_S R_D + R_G g_m R_S)}{1 + g_m R_S}, v_g = -i_t R_G, v_s = -\frac{g_m R_S i_t R_G}{1 + g_m R_S}\right]\right]$$

$$\frac{v_t}{i_t} := \frac{R_\mathrm{D} + g_m \cdot R_\mathrm{D} \cdot R_G + R_G + g_m \cdot R_S \cdot R_\mathrm{D} + R_G \cdot g_m \cdot R_S}{1 + g_m \cdot R_S} :$$

$$\frac{v_t}{i_t} := R_\mathrm{D} + R_G + \frac{g_m}{1 + g_m \cdot R_S} \cdot R_\mathrm{D} \cdot R_G :$$

P3.17

Below is the small-signal model for the circuit that we wish to analyze.



(a) For the low-frequency gain analysis we neglect all capacitances. This also means that R_G can be ignored and $v_g = v_{in}$.

$$v_{gs} = v_i - i_d \cdot R_S :$$
$$i_d = g_m \cdot v_{gs} :$$

Plugging the second equation into the first and solving for v_{gs} gives:

$$v_{gs} = \frac{v_i}{1 + g_m \cdot R_S}$$
$$i_d = g_m \cdot v_{gs} = \frac{v_i \cdot g_m}{1 + g_m \cdot R_S} = v_i \cdot G_m$$

G_m is sometimes called the compound transconductance of the transistor with source degeneration.

$$v_{out} = -i_d \cdot R_D = -v_i \cdot G_m \cdot R_D$$
$$\frac{v_{out}}{v_i} = -G_m \cdot R_D$$

(b)

$$R_G := 10[\![k\Omega]\!] : R_S := 1[\![k\Omega]\!] : R_D := 5[\![k\Omega]\!] : I_D := 500[\![uA]\!] : L := 1[\![um]\!] : W := 20[\![um]\!] :$$

$$g_m := \sqrt{2. \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot I_D} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 1.000\ [\![mS]\!]$$

$$G_m := \frac{g_m}{1 + g_m \cdot R_S} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 0.500\ [\![mS]\!]$$

$$A_{v0} := -G_m \cdot R_D = -2.500$$

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 4.067\ 10^{-14}\ [\![F]\!]$$

$$C_{gd} := W \cdot C_{ov} = 1.000\ 10^{-14}\ [\![F]\!]$$

$$R_{gs} := \frac{R_G + R_S}{1 + g_m \cdot R_S} = 5.500\ [\![kOmega]\!]$$

$$R_{gd} := R_G + R_S + G_m \cdot R_G \cdot R_S = 16.000\ [\![kOmega]\!]$$

$$f_{3\,dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_{gs} \cdot C_{gs} + R_{gd} \cdot C_{gd}} = 4.148\ 10^{8} \left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

It is interesting to compare this result to the case where $R_S = 0$.

$$R_{gs} := R_G + R_S = 11\ [\![kOmega]\!]$$

$$R_{gd} := R_G + R_S + g_m \cdot R_G \cdot R_S = 21.000\ [\![kOmega]\!]$$

$$f_{3\,dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_{gs} \cdot C_{gs} + R_{gd} \cdot C_{gd}} = 2.421\ 10^{8} \left[\!\!\left[\frac{1}{s}\right]\!\!\right]$$

Note that the bandwidth of the circuit with R_S is almost twice as high. But, unfortunately this comes at a reduction of the low-frequency gain by about the same amount. R_S can thus be used to trade gain for bandwidth. Another benefit is circuit linearization (beyond the scope of this module).